

Nexierge UAT Test Plan v2

Nexierge — UAT Test Plan & QA Handbook (v2)

Document owner: QA Lead **Last updated:** 2026-06-03 **Status:** Draft for UAT sign-off **Supersedes:** UAT_TEST_PLAN.md (v1)

1. Purpose & Scope

This document is the canonical acceptance-test reference for the Nexierge mobile app (HotelOps). It defines:

- The features under test and their acceptance criteria.
- The environment, test data, and tooling required to execute UAT.
- The defect lifecycle, severity matrix, and exit criteria.
- The roles and responsibilities for sign-off.

In scope: Flutter mobile app (Android + iOS), backend integration, FCM push, Socket.IO realtime, version control, i18n. **Out of scope:** Backend-only API correctness (covered by API test suite), admin web console, infra/SRE concerns.

2. Test Environment

Item	Value
Backend environment	uat.nexierge.io (sandbox)
Socket endpoint	wss://uat.nexierge.io/realtime
FCM project	nexierge-uat
App build channel	uat flavor, signed UAT keystore (Android) / TestFlight (iOS)
Min Android	API 24 (Android 7)
Min iOS	iOS 14
Required devices	1× low-end Android (e.g. Pixel 4a), 1× recent Android (Pixel 8 / Samsung S23), 1× iPhone (XR or later), 1× iPad (optional)
Network conditions	WiFi, 4G, 3G-throttled, Offline (via Charles / Network Link Conditioner)
Locales to verify	en, es (+ any locale active in app_localizations)

2.1 Test Data Setup

Before UAT begins, the QA lead must seed:

- **Users:** at least one per role (FrontDesk, Housekeeping, Maintenance, F&B, Manager/Admin).
- **Rooms:** ≥20 rooms across multiple floors; include OOO/OOS states.
- **Tickets:** seed open, in-progress, completed, and overdue tickets across departments.
- **Catalog:** at least 5 priced catalog items + 3 universal items.
- **Notifications:** trigger a backlog of ≥10 unread notifications per test user.

- **Version control:** server-flagged "optional" and "forced" update payloads ready to toggle.

Credentials are stored in 1Password vault **Nexierge-UAT** (request access from QA lead).

3. Roles & Responsibilities (RACI)

Activity	QA Lead	QA Tester	Dev	PM	Stakeholder
Test plan ownership	R/A	C	C	I	I
Test execution	A	R	I	I	I
Defect triage	R	C	A	C	I
Severity assignment	R/A	C	C	I	I
Sign-off	C	I	C	R	A

R = Responsible, A = Accountable, C = Consulted, I = Informed.

4. Entry Criteria

UAT begins only when **all** of the following are true:

1. Latest UAT build deployed to TestFlight / Firebase App Distribution.
 2. Backend UAT environment healthy (smoke test green).
 3. Test data seeded (§2.1).
 4. Known critical (P0/P1) defects from prior cycle are closed or have approved waivers.
 5. Release notes published, listing changes since last cycle.
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5. Exit Criteria

UAT is **passed** when:

- 100% of P0/P1 test cases pass.
 - ≥95% of P2 test cases pass; any failures have agreed mitigation.
 - Zero open P0/P1 defects.
 - ≤3 open P2 defects, each with an agreed workaround or scheduled fix.
 - Performance benchmarks (§16) within thresholds on at least one Android + one iOS device.
 - Sign-off captured from PM and Stakeholder (§22).
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6. Defect Severity Matrix

Severity	Definition	Example	Target fix
P0 — Critical	App unusable, data loss, security breach, crash on launch	Cannot log in; tickets lost on save	Same day
P1 — High	Major feature broken, no workaround	Push notifications not delivered; cannot create ticket	Within cycle
P2 — Medium	Feature degraded, workaround exists	Filter resets on tab switch; minor i18n gap	Next cycle

Severity	Definition	Example	Target fix
P3 — Low	Cosmetic, edge case	Misaligned icon at certain zoom	Backlog

6.1 Defect Report Template

ID: DEF-YYYYMMDD-###
 Title: <one-line summary>
 Severity: P0 | P1 | P2 | P3
 Module: Auth | Dashboard | Tickets | ...
 Build: <version + buildNumber>
 Device/OS: <model, OS version>
 Network: WiFi | 4G | Offline
 Locale: en | es | ...
 Steps: 1. ... 2. ... 3. ...
 Expected: <what should happen>
 Actual: <what happened>
 Frequency: Always | Intermittent | Once
 Attachments: <screenshot/video/log refs>
 Logs: <crashlytics ID / log excerpt>
 Notes: <regression? blocking other tests?>

7. Test Case Conventions

- **ID format:** <MODULE>--<NN> (e.g., TICK-05). Stable across cycles.
 - **Priority column** added to every test case: **P0** (must-pass), **P1** (must-pass for release), **P2** (should-pass), **P3** (nice-to-have).
 - Each case records: scenario, steps, expected outcome, priority, and notes.
 - Pass/Fail captured per device + locale in the execution sheet.
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8. Authentication (Auth)

ID	Priority	Scenario	Steps	Expected Outcome	Notes
AUTH-01	P0	Email/password login success	Open app → enter valid email → password → submit	Login succeeds, navigates to dashboard	Clean install + post-logout
AUTH-02	P0	Employee code login success	Toggle mode → enter employee code + login code → submit	Login succeeds	Verify toggle retains form state
AUTH-03	P0	Invalid credentials	Submit wrong creds	Inline error, login blocked	Should not navigate
AUTH-04	P1	Empty required fields	Blank fields → submit	Validation per field; no network call	
AUTH-05	P1	Network failure during login	Disable network → submit valid creds	Friendly error, no crash; retry available	
AUTH-06	P0	Session persistence	Login → kill → relaunch	User remains logged in, routes to HomeShell	
AUTH-07	P0	Logout	Profile → logout → confirm	Session cleared, login screen; back-stack not re-entered	
AUTH-08	P1	Device token registration	Login → inspect token state	FCM token registered/persisted, doesn't block login	

ID	Priority	Scenario	Steps	Expected Outcome	Notes
AUTH-09	P1	Token expiry mid-session	Force-expire token server-side, perform protected action	App refreshes silently or prompts re-login; no silent failure	New
AUTH-10	P1	Concurrent login on second device	Log in on Device B with same account	Define behaviour: allowed concurrent OR Device A invalidated. Confirm matches spec	New
AUTH-11	P2	Deep link while logged out	Open ticket deep link with no session	Routes to login; after login returns to original target	New
AUTH-12	P2	Biometric/quick re-auth (if enabled)	Enable biometric → relaunch	Prompt appears; fallback to password works	New — skip if feature not shipped
AUTH-13	P2	Brute force / rate limit feedback	Submit wrong creds repeatedly	Server-side throttle reflected in UI (e.g., "try again in N s")	New
AUTH-14	P2	Login with leading/trailing whitespace	Paste email with spaces	Trimmed and accepted, or clear validation message	New

9. Dashboard

ID	Priority	Scenario	Steps	Expected Outcome	Notes
DASH-01	P0	Bootstrap loading	Login → first launch	Shimmer until data loads; no blank content	
DASH-02	P0	KPI counts accuracy	Dashboard loads	Four KPI values match API response	Cross-check API
DASH-03	P1	Needs Attention list	Dashboard loads	High-priority tickets shown; tap → correct detail	
DASH-04	P0	Real-time KPI update	Trigger ticket change via socket	Counts update without manual refresh	
DASH-05	P1	Empty state	Zero counts	Graceful empty UI, no crash	
DASH-06	P1	KPI navigation	Tap KPI card	Routes to filtered ticket list; back returns to dashboard	
DASH-07	P1	Pull-to-refresh	Swipe down	Triggers refetch; spinner shown; data updates	New
DASH-08	P2	Stale data indicator after reconnect	Lose connection → regain	Indicator shows refresh; reconciliation runs	New
DASH-09	P1	Performance on cold load	Launch with realistic data	First paint < 1s on mid-tier device	See §16
DASH-10	P2	RBAC visibility	Login as restricted role	Only role-permitted KPIs visible	New

10. Tickets

ID	Priority	Scenario	Steps	Expected Outcome	Notes
TICK-01	P0	List load	Open tickets tab	List loads; no blank screen	

ID	Priority	Scenario	Steps	Expected Outcome	Notes
TICK-02	P1	Filter by status/scope	Switch tabs	Correct tickets per scope	Tab highlight
TICK-03	P1	Search by guest/room	Enter query	Debounced filtered results	
TICK-04	P0	Detail view	Tap ticket	All metadata, timeline, guest info correct	
TICK-05	P0	Accept incoming + ETA	Open incoming → accept w/ ETA	Status → accepted/in-progress; timeline entry added	
TICK-06	P0	Mark done	Mark as done	Ticket completes; list updates	
TICK-07	P1	Add note	Submit note	Appears in timeline; no duplicates	
TICK-08	P1	Reassign	Change department	Department + assignee updated	
TICK-09	P0	Create universal	Universal flow → submit	Created and appears in list	Required-field enforcement
TICK-10	P0	Create catalog	Catalog flow w/ pricing	Created; pricing matches selection	
TICK-11	P0	Create manual	Manual flow → submit	Created	
TICK-12	P1	Missing required data	Blank submit	Specific per-field errors; no create call	
TICK-13	P1	API failure on create	Force 500	Error UI; retry works; no orphan ticket	
TICK-14	P1	Realtime ticket update	Trigger external change	List updates; no duplicates or stale data	
TICK-15	P1	Long list performance	Scroll 200+ tickets	Smooth scroll; lazy load functional	See §16
TICK-16	P1	Draft persistence on create	Start create → background app 30s → resume	Draft preserved (or explicit discard prompt)	New
TICK-17	P2	Attachments / images	Attach image to ticket	Uploads with progress; size limit enforced; thumbnail in detail	New — verify feature exists first
TICK-18	P2	Priority / SLA badge	Open ticket near SLA breach	Priority and SLA timer displayed; escalates correctly	New
TICK-19	P2	Escalation rules	Wait past SLA threshold	Ticket auto-escalates per backend rule; UI reflects new state	New
TICK-20	P1	Concurrent edit conflict	User A and User B edit same ticket	Last-write-wins or conflict toast, per spec; no silent data loss	New
TICK-21	P1	Offline create then reconnect	Create ticket offline (if supported)	Either disabled offline OR queued + synced on reconnect	New — confirm policy
TICK-22	P2	RBAC on actions	Login as restricted role	Restricted actions (reassign, mark done) hidden/disabled	New
TICK-23	P2	PII redaction	Inspect logs / screenshots in dev	No guest PII written to logs / crash reports	New

ID	Priority	Scenario	Steps	Expected Outcome	Notes
mode					
11. Notifications (In-App Inbox)					
ID	Priority	Scenario	Steps	Expected Outcome	Notes
NOTIF-01	P0	Open inbox	Tap bell	Sheet opens with unread count	Scrollable
NOTIF-02	P0	Foreground push	Send FCM while open	Foreground toast + inbox updates	
NOTIF-03	P0	Background push	Send FCM while backgrounded	System tray entry; tap deep-links correctly	
NOTIF-04	P1	Mark read	Mark item read	Item updates; badge decreases	
NOTIF-05	P1	Clear all	Use clear-all	Empties or empty state shown	BE/UI sync
NOTIF-06	P0	Deep link tap	Tap notification	Lands on correct ticket/page with context	
NOTIF-07	P2	Offline inbox	Open with no network	Cached items shown; graceful error otherwise	
NOTIF-08	P1	Unread badge realtime	Push arrives	Bell badge updates live	
NOTIF-09	P1	Reconciliation on resume	Background app → resume	Inbox reconciles missed events (no gaps, no dupes)	New
NOTIF-10	P2	Notification grouping / channels	Send multiple notifications fast	Grouped sensibly on Android channels / iOS threading	New
NOTIF-11	P2	Notification sound + vibration per channel	Trigger each category	Matches platform notification settings	New
NOTIF-12	P2	Quiet hours / DND respect	Enable DND	Pushes respect OS DND; in-app inbox still updates	New

12. Profile

ID	Priority	Scenario	Steps	Expected Outcome	Notes
PROF-01	P0	View profile	Open profile	Name, email, hotel, subscription correct	
PROF-02	P1	Edit name	Edit → save	Persists locally + server	
PROF-03	P1	Avatar from gallery	Upload	Progress shown; new avatar visible	
PROF-04	P1	Avatar from camera	Upload	Permission prompt; upload works	
PROF-05	P0	Theme toggle	Toggle light/dark	Switches instantly; persisted (app.themeMode)	
PROF-06	P0	Language change	Pick locale	UI updates immediately; persisted (app.locale)	

ID	Priority	Scenario	Steps	Expected Outcome	Notes
PROF-07	P0	Logout	Logout + confirm	Session cleared	
PROF-08	P2	Invalid edit data	Bad input	Validation blocks request	
PROF-09	P1	Profile API failure	Force failure	Error + retry path	
PROF-10	P2	Avatar size/format limits	Upload >max size / unsupported format	Clear error; no crash	New
PROF-11	P2	Theme/locale survives logout-login	Toggle → logout → login	Restored on next session	New

13. Activity Feed

ID	Priority	Scenario	Steps	Expected Outcome	Notes
ACT-01	P0	Timeline load	Open tab	Events load, grouped by day	
ACT-02	P1	Event-type filter	Select	Only matching events	Persists while visible
ACT-03	P1	Department filter	Apply	Only relevant events	Combine filters supported
ACT-04	P0	Navigate from event	Tap row	Opens ticket detail; message if missing	
ACT-05	P1	Realtime event	Trigger new event	Appears live, no dupes	
ACT-06	P2	Empty state	No events	Clear empty UI	
ACT-07	P1	Scroll performance	Scroll fast	Smooth on long feed	See §16
ACT-08	P2	Timezone correctness	Change device TZ	Timestamps render in correct local TZ	New
ACT-09	P2	Date format per locale	Switch locale	Date/number formatting respects locale	New

14. Rooms / Catalog / FCM / Languages / Modules / Version / Shell

Cases from v1 retained; additions below.

14.1 Rooms

ID	Priority	Scenario	Notes
ROOM-01	P1	Room selection in ticket creation	v1
ROOM-02	P1	Room context in tickets	v1
ROOM-03	P2	No-room state	v1
ROOM-04	P2	Room OOO/OOS state visibility	New

14.2 FCM / Push

ID	Priority	Scenario	Notes
FCM-01	P0	Initial token registration	v1
FCM-02	P1	Token refresh	v1
FCM-03	P0	iOS/Android permission flow	v1
FCM-04	P0	Payload routing	v1
FCM-05	P0	Terminated-state launch via push	v1
FCM-06	P2	Notification permission revoked at runtime	New — surface in-app prompt + settings link
FCM-07	P2	Token rotation after app reinstall	New

14.3 Languages / i18n

ID	Priority	Scenario	Notes
LANG-01	P0	Change locale in profile	v1
LANG-02	P0	Locale persists across restart	v1; key <code>app.locale</code>
LANG-03	P1	Fallback to system locale	v1
LANG-04	P1	Translation coverage	v1; sweep ARB keys
LANG-05	P2	RTL layout (if supported)	New
LANG-06	P1	No hardcoded strings (UI/snackbars/dialogs/notifications)	Per CLAUDE.md i18n rule

14.4 Modules placeholder

ID	Priority	Scenario	Notes
MOD-01	P2	Coming-soon screen	v1
MOD-02	P3	Placeholder under rotation	v1

14.5 Version Control

ID	Priority	Scenario	Notes
VER-01	P1	Optional update prompt	v1
VER-02	P0	Forced update gating	v1
VER-03	P1	Up-to-date path	v1
VER-04	P1	Version check once per session	v1
VER-05	P2	App store link opens correct store per platform	New

14.6 Shell / Navigation

ID	Priority	Scenario	Notes
SHELL-01	P0	Bottom nav switching	v1
SHELL-02	P0	Deep link routing (cold, warm, terminated)	v1 + verify all three states
SHELL-03	P1	Back stack across tabs	v1
SHELL-04	P1	HomeShell resume stability	v1
SHELL-05	P2	Tablet / landscape layout	New
SHELL-06	P2	Split-screen / multi-window (Android)	New

15. Realtime / Socket Resilience

ID	Priority	Scenario	Expected Outcome
RT-01	P0	Initial socket connect on login	Connection established; auth handshake succeeds
RT-02	P0	Reconnect after network drop	Exponential backoff; reconnects without manual action
RT-03	P0	Reconcile-on-reconnect	Missed events fetched via REST diff (no UI gaps, no dupes)
RT-04	P1	Reconcile on app resume from background	Same as RT-03
RT-05	P1	Heartbeat / ping behaviour	No false disconnects on idle
RT-06	P2	Periodic poll fallback (last resort)	Only engages if socket unhealthy beyond threshold
RT-07	P1	Auth-failure on socket (expired token)	Triggers token refresh and reconnect

16. Performance & Stability Benchmarks

ID	Priority	Metric	Target	Method
PERF-01	P1	Cold start to first interactable frame	≤ 2.5s mid-tier Android, ≤ 2s iPhone	Stopwatch + Flutter DevTools
PERF-02	P1	Dashboard first paint after auth	≤ 1s with cached data, ≤ 2s cold	DevTools timeline
PERF-03	P1	Screen transition jank	< 1% frames > 16ms	DevTools
PERF-04	P2	Memory after 30 min usage	No upward leak trend	DevTools memory tab
PERF-05	P1	Network loss resilience	Graceful errors; recovery on reconnect	Charles offline toggle
PERF-06	P1	API 5xx handling	Friendly error + retry on critical calls	Charles rewrite rules
PERF-07	P2	Offline cache fallback	Stale-data indicator shown	
PERF-08	P0	No crashes during smoke run	Zero crashes across full P0 pass	Crashlytics
PERF-09	P0	Session expiry	Clean re-login prompt, no unauthorised access	
PERF-10	P1	Localization completeness	No missing strings; no overflow in es	Visual sweep + ARB diff

17. Accessibility (A11y)

ID	Priority	Scenario	Expected Outcome
A11Y-01	P1	TalkBack / VoiceOver labels on primary actions	All interactive elements announced meaningfully
A11Y-02	P1	Dynamic text scale 130%	Layouts adapt without clipping/overflow
A11Y-03	P2	Color contrast (WCAG AA)	Text vs background passes AA
A11Y-04	P2	Focus order with external keyboard (iPad)	Logical focus traversal
A11Y-05	P2	Tap target size $\geq 44 \times 44$ pt	Verified on key actions

18. Security & Privacy

ID	Priority	Scenario	Expected Outcome
SEC-01	P0	Tokens stored in secure storage (Keychain/Keystore)	Not in shared prefs / logs
SEC-02	P0	TLS only; no cleartext traffic	Verified via networkSecurityConfig / ATS
SEC-03	P1	Auth headers redacted in logs	No bearer tokens in Crashlytics/logcat
SEC-04	P1	Deep link target validation	Malformed / external links rejected
SEC-05	P1	Backgrounded app blurs sensitive screens (iOS app switcher)	If feature shipped, verify; else flag as known gap
SEC-06	P2	Clipboard hygiene for guest PII	Sensitive fields don't auto-copy
SEC-07	P1	Logout invalidates server session	Old token rejected on next call
SEC-08	P1	Crash reports contain no PII	Manual inspection of recent Crashlytics events

19. Observability

ID	Priority	Scenario	Expected Outcome
OBS-01	P1	Crashlytics receives crashes	Force-crash dev build; entry appears
OBS-02	P2	Analytics events fire for key actions	login, ticket_create, ticket_complete, push_open
OBS-03	P2	Performance traces (cold start, ticket list)	Visible in Firebase Performance
OBS-04	P2	Structured logs (no PII)	Log review on a sample session

20. Regression Suite (Smoke — must pass every build)

Run this short list on every build before deeper UAT:

AUTH-01, AUTH-06, AUTH-07, DASH-01, DASH-02, DASH-04, TICK-01, TICK-04, TICK-05, TICK-06, TICK-09, NOTIF-01, NOTIF-03, NOTIF-06, PROF-05, PROF-06, FCM-01, FCM-04, LANG-01, VER-02, SHELL-01, RT-01, RT-03, PERF-08.

Smoke target: complete in ≤ 60 minutes on one Android + one iOS device.

21. Traceability Matrix (Requirement → Test)

Requirement	Covered by
R-AUTH: Secure login + session persistence	AUTH-01..14, SEC-01, SEC-07
R-DASH: Live KPIs + needs attention	DASH-01..10, RT-04
R-TICK: Full ticket lifecycle	TICK-01..23, RT-03
R-NOTIF: Inbox + push	NOTIF-01..12, FCM-01..07
R-PROF: Profile + preferences	PROF-01..11, LANG-01..06
R-ACT: Activity feed	ACT-01..09
R-VER: Update gating	VER-01..05
R-A11Y: Accessibility baseline	A11Y-01..05
R-SEC: Security & privacy	SEC-01..08
R-OBS: Observability	OBS-01..04

Update this table whenever a requirement is added or retired.

22. Sign-off

UAT is complete when the following sign off in writing (email or PR comment):

Role	Name	Date	Decision
QA Lead			Pass / Fail
Dev Lead			Pass / Fail
PM			Pass / Fail
Stakeholder			Pass / Fail

Attach: execution sheet (per-case Pass/Fail per device/locale), open-defect list with severity & owner, performance benchmark capture.

23. Gaps & Open Questions (to confirm before UAT starts)

The following items in v1 were ambiguous; confirm with product before execution:

1. **Concurrent login policy** (AUTH-10) — allow or invalidate?
2. **Offline ticket create** (TICK-21) — disabled or queued?
3. **Concurrent edit conflict** (TICK-20) — last-write-wins, lock, or merge?
4. **Attachments** (TICK-17) — supported in this release?
5. **Biometric auth** (AUTH-12) — in scope?
6. **RTL support** (LANG-05) — required for current locales?
7. **Backgrounded-screen blur for PII** (SEC-05) — implemented?

24. Change Log

Version	Date	Author	Notes
v1	(existing)	—	Initial test plan in UAT_TEST_PLAN.md
v2	2026-06-03	QA Lead	Expanded to full QA handbook: env, RACI, severity, exit criteria, traceability, accessibility, security, observability, realtime resilience, and ~40 new test cases

Appendix A — Converting this document to PDF

Use any of the methods in CONVERT_TO_PDF.md. Quickest:

```
# From repo root
pandoc docs/UAT_TEST_PLAN_v2.md \
  -o docs/UAT_TEST_PLAN_v2.pdf \
  --from gfm \
  --pdf-engine=xelatex \
  -V geometry:margin=0.7in \
  -V mainfont="Helvetica"
```

If pandoc / LaTeX is unavailable, use VS Code's **Markdown PDF** extension (right-click → Export PDF) or the Node script at scripts/.